

Data Visualization - Sales Dashboard

Project Title: Data Visualization - Sales Dashboard

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Subject: Data Visualization | Year: 2025-26

1. Objective:

To create a sales dashboard using data visualization techniques to analyze city-wise and product-wise sales performance.

2. Dataset Used:

Products: Laptop, Mobile, Mouse, Keyboard, Charger

Price Range: 2000 to 50000 | Sales Qty: 150 to 400

Cities: Hyderabad, Bangalore, Chennai, Delhi | 5 Rows

3. Tools:

Python, Pandas, Matplotlib, Seaborn for visualization

4. Visualization Steps:

Step 1: Data collection and cleaning

Step 2: City-wise sales aggregation

Step 3: Create 4 charts - Bar, Pie, Scatter, Bar

Step 4: Build dashboard layout

Step 5: Generate insights for business decisions

Data Visualization - Code

5. Source Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = [
    {'Product': 'Laptop', 'Price': 50000, 'City': 'Hyderabad', 'Sales': 350},
    {'Product': 'Mobile', 'Price': 20000, 'City': 'Bangalore', 'Sales': 200},
    {'Product': 'Mouse', 'Price': 5000, 'City': 'Hyderabad', 'Sales': 400},
    {'Product': 'Keyboard', 'Price': 30000, 'City': 'Chennai', 'Sales': 150},
    {'Product': 'Charger', 'Price': 2000, 'City': 'Delhi', 'Sales': 300}
]
df = pd.DataFrame(data)

# Dashboard - 4 Visualizations
plt.figure(figsize=(12,8))

# Chart 1: City Sales
plt.subplot(2,2,1)
df.groupby('City')['Sales'].sum().plot(kind='bar', color='teal')
plt.title('Sales by City')

# Chart 2: Price Share
plt.subplot(2,2,2)
plt.pie(df['Price'], labels=df['Product'], autopct='%1.1f%%')
plt.title('Price Share')

# Chart 3: Price vs Sales
plt.subplot(2,2,3)
plt.scatter(df['Price'], df['Sales'])
plt.title('Price vs Sales')

# Chart 4: Product Sales
plt.subplot(2,2,4)
plt.bar(df['Product'], df['Sales'], color='orange')
plt.title('Product-wise Sales')

plt.tight_layout()
plt.savefig('sales_dashboard.png', dpi=300)
plt.show()
```

Data Visualization - Results

6. Observations from Dashboard:

1. City Performance: Hyderabad top with 750 total sales (2 products).
2. Product Sales: Mouse highest qty 400, Keyboard lowest 150.
3. Price Analysis: Laptop highest price 50000 dominates value.
4. Price vs Sales: Low price products have high sales quantity.

7. Conclusion:

Data visualization dashboard shows Hyderabad is best market. Budget products sell more in quantity. Premium products give more revenue.

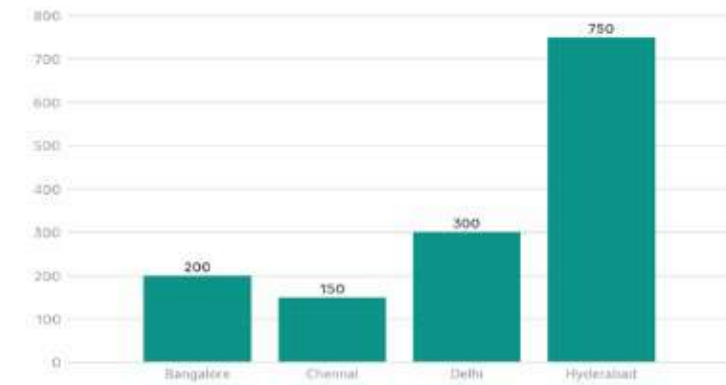
8. Business Recommendations:

Focus marketing in Hyderabad. Increase stock of Mouse and Charger. Launch offer on Laptop for higher revenue growth.

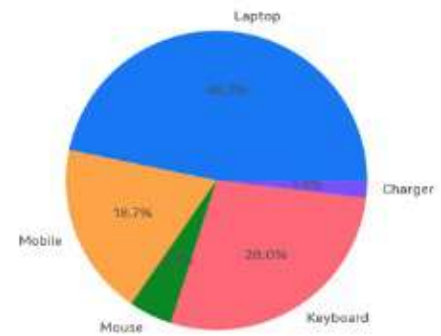
Data Visualization - Charts Output

Data Visualization - Sales Dashboard - 4 Charts

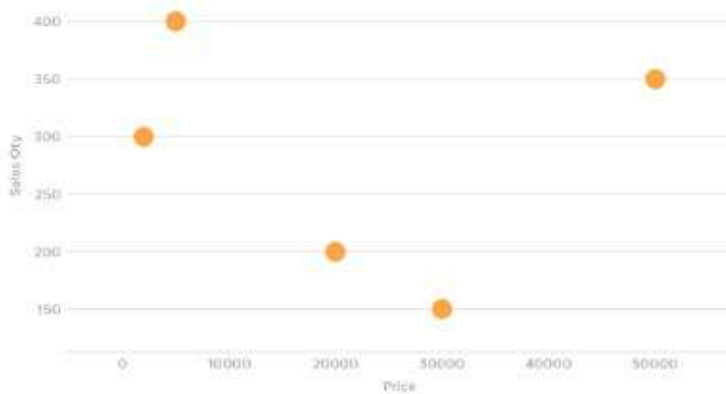
1. Sales by City



2. Price Share by Product



3. Price vs Sales



4. Product-wise Sales

